Week 2 Assignment: Welcome to the World of Objects

Business Case Options (with Descriptions):

1. **Library Management System** **<u>(Example for inspiration, do NOT pick this case</u>)**:
    - **Description**: This system manages books, students (members), librarians, and borrowing activities in a university library. Students can search for available books, borrow them, and return them. Librarians can manage book inventory and handle borrowing requests.
    - **Glossary Example**: Book, Student, Librarian, Borrowing, Reservation.
    - **Use Cases**: Borrow Book, Return Book, Search Catalog, Reserve Book.
    - **Domain Objects**: Book, Student, Librarian, Borrowing Transaction.
2. **Online Food Ordering System**:
    - **Description**: This system allows customers to browse menus from various restaurants, place orders, and track deliveries. Restaurant owners can manage their menus, and delivery staff can update the status of deliveries.
3. **Student Course Registration System**:
    - **Description**: This system allows students to register for courses, view class schedules, and manage academic plans. Professors can manage course offerings and view enrolment lists.
4. **Parking Management System**:
    - **Description**: This system manages parking spaces for an office complex. Employees can reserve parking spots, check availability, and manage their reservations. Administrators can manage parking assignments and ensure efficient use of spaces.
5. **Online Event Ticketing System**:
    - **Description**: This platform allows users to purchase tickets for events such as concerts, sports games, and theatre shows. Event organisers can list events, and users can view event details and book tickets.

Part 1: Group Task – Glossary, Object-Oriented Analysis, Use Case Diagram, and Use Case Descriptions

Task:

In your group (2 - 4 students maximum), choose one of the business cases provided (**Not Library**). You will first create a **Glossary** to define key terms, then perform **Object-Oriented Analysis (OOA)**, create a **Use Case Diagram**, and write **Use Case Descriptions** to represent the business case.

Steps:

1. **Create a Glossary**:
    - Define the key terms related to your system.
    - **Example (Library System)**:
        - **Book**: A physical or digital item available for borrowing.
        - **Borrowing**: The process of taking out a book for a period of time.
        - **Librarian**: A person responsible for managing library resources and assisting students with their library needs.
        - **Student**: A member of the library who can borrow books.
        - **Reservation**: An action to request a book that is currently unavailable for future borrowing.
        - …
2. **Object-Oriented Analysis (OOA)**:
    - Identify the core objects and describe their responsibilities.
    - Example (Library Management System):
      Objects can be: Book, Student, Library employee, Loan, Reservation
        - Book has properties such as title, author, availability.
        - Student can borrow and reserve books.
        - Loan registers which student has which book at what time.
3. **Functional & Non-functional Requirements (FURPS)**:
    - Draw up the requirements for your system according to the FURPS model. Formulate the functional requirements using the two methods we saw in class: in the form of user stories and in the form of use case descriptions.
    - Example (Library Management System):
        - Functional requirements – User Story
          As a student, I want to be able to search for books by title or author so that I can quickly find what I need.
        - Functional requirements – Use Case Description
          Title: Borrow Book.
          Actors: Student, Librarian.
          Preconditions: The student must have a valid library card, and the book must be available.
          Normal Flow: The student requests to borrow a book, the librarian checks the book's availability, and the librarian processes the borrowing transaction.

Alternate Flow: If the book is not available, the student is notified and can reserve the book for a future date.
Postconditions: The book is lent to the student, and the borrowing record is updated.

- Non-functional requirements (URPS):
  Usability: The search function must be intuitive and work via a simple search bar.
  Reliability: The system must be available with an uptime of at least 99.5%.
  Performance: Search results must be displayed within 2 seconds.
  Supportability: The system must keep log files of search queries for error analysis.

Deliverables:

1. **Group Deliverable**: Complete the following components based on the chosen business case:
   a. **Glossary:** A document containing definitions of the most important terms within the chosen system.
   b. **Object-Oriented Analysis (OOA):** Identification and description of the core objects and their responsibilities within the system.
   c. **Requirements (FURPS model):**
      i. Formulate non-functional requirements for: **Usability, Reliability, Performance and Supportability.**
      ii. Formulate the functional requirements in **two ways:**
         1. **User stories**
         2. **Use case descriptions** (use the template)

Part 2: Individual Research Task – Use case Diagram

Assignment (for next class):

In preparation, each student will conduct individual research on **Use Case Diagrams**.

Answer in a short summary:

- What is a Use Case Diagram?
- What components does it contain (actors, use cases, relationships)?
- What is its purpose in software development?

Application:

Draw your own **Use Case Diagram** based on the case your group has chosen.

Use your research as supporting evidence when drawing the diagram.

To be submitted individually:

- A short summary of your findings (±200 words).
- A drawn Use Case Diagram (digital).

Grading Criteria:

- **Group work:**
  - **Glossary:** Are key terms defined correctly and completely?
  - **OOA:** Are core objects chosen logically and well developed?
  - **Requirements**: Are functional and non-functional requirements clear and applicable?
  - **Use Case Descriptions**: Are the descriptions complete, clear and realistic?
- **Individual work**:
  - **Use Case Diagram Research**: Does the student understand the structure and function of the diagram?
  - **Drawing**: Does the diagram contain the correct components and is it applicable to the chosen case?